

tf.compat.v1.nn.max_pool_with_argmax

Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/max_pool_with_argmax

Performs max pooling on the input and outputs both max values and indices.

Function Signatures

```
tf.compat.v1.nn.max_pool_with_argmax( input, ksize, strides,
padding, data_format='NHWC', Targmax=None, name=None,
output_dtype=None, include_batch_in_index=False )
(y * width + x) * channels + c
```

Code Examples

```
tf.compat.v1.nn.max_pool_with_argmax(  
    input,  
    ksize,  
    strides,  
    padding,  
    data_format='NHWC',  
    Targmax=None,  
    name=None,  
    output_dtype=None,  
    include_batch_in_index=False  
)
```

```
tf.compat.v1.nn.max_pool_with_argmax(  
    input,  
    ksize,  
    strides,  
    padding,  
    data_format='NHWC',  
    Targmax=None,  
    name=None,  
    output_dtype=None,  
    include_batch_in_index=False  
)
```

Documentation

Performs max pooling on the input and outputs both max values and indices.

The indices in `argmax` are flattened, so that a maximum value at position `[b, y, x, c]` becomes flattened index:

$(y * \text{width} + x) * \text{channels} + c$ if `include_batch_in_index` is `False`;

$((b * \text{height} + y) * \text{width} + x) * \text{channels} + c$ if `include_batch_in_index` is `True`.

The indices returned are always in $[0, \text{height}) \times [0, \text{width})$ before flattening, even if padding is involved and the mathematically correct answer is outside (either negative or too large). This is a bug, but fixing it is difficult to do in a safe backwards compatible way, especially due to flattening.

Returns